Teacher guide and distribution order

1. Give A first. Preserve the original hypothesis, reason and design before giving B.
2. Complete B in order: main inquiry, tip comparison, then agar（瓊脂） inquiry. Preserve each prediction and reason before showing its results. Cover results not yet reached.
3. Fix the answers to all three inquiries before giving C. Keep D and the complete teacher pack for teachers only.
4. Keep original answers and record revisions separately. Do not convert time, clicks or completion into ability scores.
5. Static diagrams use the VL teaching model, not real experimental data. Students interpret directions and measure angles themselves. Do not add calculations for quantities not measured in the VL.

Research procedure alignment

Under proposal 4.1 / 4.3.2, both groups use identical scenarios, concepts and questions before formal teaching. Keep four-week activity worksheets separate from the 10-item SPSAI and 4-item CKT pre/post-tests. Do not add experimental-group opinion questions or interviews.

Paper learners read static data and use a protractor; VL learners use animation and an adjustable pointer. These operations are not identical. The researcher must check equal time and data access, original-answer copies, scoring consistency and comparability of skill evidence.

Teacher check: evidence limits

The tip and agar（瓊脂） models do not identify substance X on their own, nor directly measure auxin（生長素） distribution under light from one side. These limits belong to the teacher guide; removed student questions are not restored.

Independent variable	Light-blocking treatment
Dependent variable	The coleoptile（胚芽鞘）’s bending response
Controlled variables	Light direction and intensity; Coleoptile（胚芽鞘） type, starting height and growth stage; Time exposed to light; Temperature and water supply

Experiment assumptions

- The coleoptiles（胚芽鞘） start in similar growth conditions.
- The light-blocking cap and sleeve do not limit elongation（延長） or bending.
- The treatments do not cause clear temperature differences between setups.

A · No light blocked	Bends left
B · Light blocked at the tip	No clear bending
C · Light blocked below the tip	Bends left
A · No light blocked · Elongation（延長）	Yes
D · Tip removed · Elongation（延長）	No
E · Blank agar（瓊脂） · Centre	0°（Teaching model）; No clear bending
F · Treated agar（瓊脂） · Centre	0°（Teaching model）; No clear bending
G · Treated agar（瓊脂） · Left	35°（Teaching model; ±2°）; Bends right
H · Treated agar（瓊脂） · Right	35°（Teaching model; ±2°）; Bends left

3. Which part may sense light from one side?	Tip
1. Compare A and B: What changes when light is blocked at the tip?	No clear bending towards the light.
2. Compare A and C: Does it still bend when the lower part receives no direct light?	It can still bend towards light.
1. What does comparing elongation（延長）in A and D support?	The presence of the tip also affects the coleoptile（胚芽鞘）' s growth / elongation（延長）.
1. What does the comparison of E and F support?	The tip may produce a growth-promoting effect that can be transferred.
2. What does the comparison of F, G and H support?	The position of treated agar（瓊脂） affects the direction of coleoptile（胚芽鞘） bending.

3. When treated agar（瓊脂） is placed on the **left** side of a coleoptile（胚芽鞘） with its tip removed, compared with the right side, the left side receives more substance X, causing cells on the left to **elongate（延長）** more and the coleoptile（胚芽鞘） to bend to the right side.

Learning reflection

The light source is on the right, and the coleoptile（胚芽鞘） tip senses light from one side. Auxin（生長素） moves towards the left (shaded side) and passes downwards, causing cells on the left to elongate（延長） more. The coleoptile（胚芽鞘） therefore bends towards the light on the right, showing positive phototropism（向光性）. The teacher reviews this; it is not automatically marked wrong.

Assessment item	Maximum score	Criteria / notes
Cross-VL calibration status		This draft rubric has not been validated or calibrated across VLs. Do not convert time, completion rate or clicks directly into ability scores.
First observations	2	2: accurately describes the coleoptile（胚芽鞘）, light from one side and bending; 1: partial description; 0: no relevant description.
Qualitative observations	2	The bending directions of A, B and C in the main inquiry total 1 mark; the elongation（延長） observations of A and D in extension 1 total 1 mark. Score older records using their original questions and models. Preserve original observations.
Variable classification	4	Independent and dependent variables: 1 each. Controlled variables: 2 for all correct choices with none extra; 0 for a wrong or missing choice.
Original hypothesis and reasons	2	2: testable predictions and reasonable reasons for all three stages; 1: partly complete; 0: no testable content. Do not deduct marks just because a prediction differs from the model. Score older records using their original inquiry.
Experiment assumptions	1	1: select similar initial conditions, no mechanical restriction and no clear temperature difference, without selecting that the same species needs no size control. Use the original options for older records.
Control group	1	1: compare uncovered A with B covered at the tip and C covered below the tip, keeping other conditions the same. Review the intact A versus cut-tip D comparison in extension 1 as additional evidence. Use the original design for older records.
Experimental operation	2	Extension 2: accept only 0° for E and F and 33–37° (35° ±2°) for G and H; 0.5 per setup. The teacher still assesses the measurement method and setup quality. Do not score clicks or time.
Experiment setup	2	2: clearly shows the light-blocking treatments for A, B and C, light from one side and controlled conditions; 1: partly complete; 0: no valid comparison. Use the original design for older records.
Comparison conclusions	2	The 3 main questions, 1 extension 1 question and 3 extension 2 questions total 2 marks, proportional to correct answers. Each blank in extension 2 question 3 is one third of that question; its total weight equals any other comparison question. Use original questions for older records.
Data interpretation	2	Assess the learning reflection: 2 for linking light on the right, sensing at the tip, more auxin（生長素） on the left (shaded side) passing down, more cell elongation（延長） on the left and bending right; 1 for partial links; 0 for no relevant explanation. The teacher assesses this, without automatic keyword scoring. Use original questions for older records.
Reflection expression	2	2: a clear, coherent cause-and-effect explanation of the coleoptile（胚芽鞘） bending right in the context diagram; 1: partly clear; 0: no relevant reflection. For older records, assess original ideas, comparison evidence and revisions.
Data recording and evidence communication	2	The teacher assesses clear communication of angles, directions and comparison evidence: 2 for clear and complete; 1 for partly clear; 0 for no relevant communication. Retain original chart scoring for older records.

Assessment item	Maximum score	Criteria / notes
Terminology and light-sensing part	2	Teacher assessment: 1 each for correct use of phototropism（向光性） / positive phototropism（向光性） and sensing light at the tip, considering inquiry answers and reflection.
Light-sensing and growing parts	2	2: correctly links unequal elongation（延長） on the two sides with bending direction, including bending left with light on the left and bending right with agar（瓊脂） on the left; 1: partly correct; 0: no demonstrated understanding.
Auxin（生長素） mechanism	2	2: explains that auxin（生長素） is a plant hormone and the tip produces a growth-promoting effect that can pass downwards; 1: partly correct; 0: no demonstrated understanding.
Context application / earlier evidence revision	2	2: correctly applies learning points to light on the right, explaining that the left is shaded and elongates（延長） more, causing bending right; 1: partly correct; 0: no relevant application. Use original questions for older records.
Total score	32	Six SPS categories, 4 each, total 24; new knowledge totals 8. Show the overall score only after all 11 teacher scores and reflection are complete (10 teacher scores for older records).
Blank versus zero		A blank means awaiting assessment; 0 is an explicit zero score. Excel recalculates formulas when opened.
Saving teacher scores		Teacher scores exist only in the saved Excel file. Downloading again does not import teacher scores from an earlier file.